

C Programming: Functions and Recursion

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Instructions

- Write a separate C program for each question.
- Use user-defined functions wherever specified.
- For recursive problems, identify the base case clearly.
- Test your program using the given sample test cases.

Topics Covered

Functions, parameters, return values, number-based problems, and recursion.

Question 1: Largest of Three Numbers



Problem

Write a C program using a user-defined function `largest()` that accepts three integers and returns the largest number.

Example:

Input: 15 42 27

Output: Largest = 42

Sample Test Cases:

Input	Expected Output
10 20 30	Largest = 30
45 12 32	Largest = 45
8 8 5	Largest = 8

Question 2: Even or Odd



Problem

Write a function `isEven()` that accepts an integer and returns 1 if the number is even and 0 otherwise. Display whether the given number is even or odd.

Example:

Input: 24

Output: 24 is Even

Sample Test Cases:

Input	Expected Output
24	Even
17	Odd
0	Even

Question 3: Positive, Negative or Zero



Problem

Write a function `checkNumber()` that accepts an integer and determines whether it is positive, negative, or zero.

Example:

Input: -15

Output: Negative

Sample Test Cases:

Input	Expected Output
25	Positive
-8	Negative
0	Zero

Question 4: Prime Number



Problem

Write a function `isPrime()` that accepts an integer and returns 1 if the number is prime and 0 otherwise.

Example:

Input: 29

Output: 29 is Prime

Sample Test Cases:

Input	Expected Output
29	Prime
15	Not Prime
2	Prime
1	Not Prime

Question 5: Prime Numbers in a Range



Problem

Write a C program using the function `isPrime()` to print all prime numbers between two given integers.

Example:

Input: 10 30

Output: 11 13 17 19 23 29

Sample Test Cases:

Input	Expected Output
1 10	2 3 5 7
10 20	11 13 17 19
20 35	23 29 31

Question 6: Factorial Using Function



Problem

Write a function `factorial()` that accepts an integer and returns its factorial.

Example:

Input: 5

Output: Factorial = 120

Sample Test Cases:

Input	Expected Output
5	120
6	720
0	1

Question 7: Power of a Number



Problem

Write a function `power()` that accepts two integers x and n and returns x^n . Do not use the library function `pow()`.

Example:

Input: 2 5

Output: 32

Sample Test Cases:

Input	Expected Output
2 5	32
3 4	81
5 0	1

Question 8: Palindrome Number



Problem

Write a function `isPalindrome()` that accepts an integer and checks whether the number is a palindrome.

Example:

Input: 1221

Output: Palindrome

Sample Test Cases:

Input	Expected Output
1221	Palindrome
1234	Not Palindrome
555	Palindrome

Question 9: Armstrong Number



Definition

An Armstrong number is a number in which the sum of its digits, each raised to the power of the total number of digits, is equal to the original number.

$$153 = 1^3 + 5^3 + 3^3 = 153$$

Therefore, 153 is an Armstrong number.

Problem

Write a function `isArmstrong()` that accepts an integer and checks whether it is an Armstrong number.

Sample Test Cases:

Input	Expected Output
153	Armstrong
370	Armstrong
123	Not Armstrong

Question 10: Count Digits



Problem

Write a function `countDigits()` that accepts an integer and returns the number of digits in the number.

Example:

Input: 12345

Output: Number of digits = 5

Sample Test Cases:

Input	Expected Output
12345	5
908	3
7	1

Question 11: Sum of Digits



Problem

Write a function `sumDigits()` that accepts an integer and returns the sum of its digits.

Example:

Input: 12345

Output: Sum = 15

Sample Test Cases:

Input	Expected Output
12345	15
908	17
555	15

Question 12: Reverse a Number



Problem

Write a function `reverseNumber()` that accepts an integer and returns its reverse.

Example:

Input: 12345

Output: Reverse = 54321

Sample Test Cases:

Input	Expected Output
12345	54321
908	809
1221	1221

Question 13: Factorial Using Recursion



Problem

Write a recursive function to calculate the factorial of a given number.

Example:

Input: 5

Output: 120

Sample Test Cases:

Input	Expected Output
5	120
4	24
0	1

Question 14: Sum of Even Numbers Using Recursion



Problem

Write a recursive function to calculate the sum of all even numbers from 1 to a given positive integer n .

Example:

Input: 10

Output: Sum = 30

Sample Test Cases:

Input	Expected Output
10	30
8	20
5	6

Question 15: Power Using Recursion



Problem

Write a recursive function to calculate x^n without using the library function `pow()`.

Example:

Input: 2 5

Output: 32

Sample Test Cases:

Input	Expected Output
2 5	32
3 3	27
7 0	1

Question 16: Sum of Digits Using Recursion



Problem

Write a recursive function to calculate the sum of the digits of a given integer.

Example:

Input: 12345

Output: Sum = 15

Sample Test Cases:

Input	Expected Output
12345	15
908	17
1111	4

Question 17: Reverse a Number Using Recursion



Problem

Write a recursive function to print the digits of a given number in reverse order.

Example:

Input: 12345

Output: 54321

Sample Test Cases:

Input	Expected Output
12345	54321
908	809
1234	4321

Question 18: Fibonacci Using Recursion



Problem

Write a recursive function to find the n -th Fibonacci number.

$$F(n) = F(n - 1) + F(n - 2)$$

Example:

Input: 7

Output: 13

Sample Test Cases:

Input	Expected Output
5	5
7	13
10	55

Question 19: Palindrome Using Recursion



Problem

Write a recursive function to check whether a given integer is a palindrome.

Example:

Input: 1221

Output: Palindrome

Sample Test Cases:

Input	Expected Output
1221	Palindrome
1234	Not Palindrome
1331	Palindrome

Question 20: Count Digits Using Recursion



Problem

Write a recursive function to find the number of digits in a given integer.

Example:

Input: 123456

Output: Number of digits = 6

Sample Test Cases:

Input	Expected Output
123456	6
908	3
7	1

Question 21: Armstrong Numbers in a Range



Problem

Write a C program using functions to print all Armstrong numbers between two given integers.

Create separate functions to:

- count the number of digits,
- calculate the required power,
- check whether a number is an Armstrong number.

Example:

Input:

1 1000

Output:

1 2 3 4 5 6 7 8 9

153 370 371 407

Sample Test Cases:

Input	Expected Output
1 500	1-9, 153, 370, 371
100 500	153, 370, 371, 407
1 1000	1-9, 153, 370, 371, 407

Question 22: Strong Number



Definition

A Strong number is a number in which the sum of the factorials of its digits is equal to the original number.

For example,

$$\begin{aligned} 145 &= 1! + 4! + 5! \\ &= 1 + 24 + 120 = 145 \end{aligned}$$

Problem

Write a C program using a function `factorial()` to determine whether a given number is a Strong number.

Example:

Input: 145

Output: Strong Number

Sample Test Cases:

Input	Expected Output
145	Strong Number
123	Not a Strong Number
40585	Strong Number

Hint

Extract each digit and calculate its factorial using the `factorial()` function.

Question 23: Strong Numbers in a Range



Problem

Write a C program using functions to display all Strong numbers between two given integers.

Use the `factorial()` function to calculate the factorial of each digit.

Approach

For every number in the range:

- 1 Extract its digits.
- 2 Find the factorial of each digit.
- 3 Add the factorials.

Example:

Input:

1 1000

Output:

1 2 145

Sample Test Cases:

Input	Expected Output
1 100	1 2
1 500	1 2 145
100 1000	145

Question 24: Decimal to Binary Using Recursion



Problem

Write a recursive function to convert a positive decimal integer into its binary representation.

Do not use arrays or library conversion functions. The recursive function should directly print the binary digits in the correct order.

Hint

Divide the number by 2 recursively. Print the remainder after returning from the recursive call.

Example:

Input: 25

Output: 11001

Sample Test Cases:

Input	Expected Output
10	1010
25	11001
32	100000

Recursive idea:

$$n \rightarrow \frac{n}{2}$$

Print the remainder while returning.

Question 25: Sum of Even and Odd Digits



Problem

Write two recursive functions: `sumEvenDigits()` and `sumOddDigits()`.

The program should read an integer and calculate the sum of its even digits and odd digits separately.

Hint

Extract one digit at a time using `% 10` and recursively process the remaining digits.

Example:

Input: 123456

Even digits: $2 + 4 + 6 = 12$

Odd digits: $1 + 3 + 5 = 9$

Output:

Sum of Even Digits = 12

Sum of Odd Digits = 9

Sample Test Cases:

Input	Even Sum	Odd Sum
123456	12	9
2468	20	0
13579	0	25

Question 26: Simple Linear Regression



Problem

Write a C program to implement simple linear regression using user-defined functions.

Given n pairs of observations (x_i, y_i) , determine the regression equation:

$$y = a + bx$$

Calculate the values of a and b , and predict the value of y for a given value of x .

Required Functions

Implement suitable user-defined functions:

- `calculateSlope()`
- `calculateIntercept()`
- `predictY()`

Input

The program should accept:

- Number of observations n
- n values of x
- n corresponding values of y
- A value of x for prediction

Question 26: Algorithm and Example



Algorithm

- 1 Read n observations (x_i, y_i) .
- 2 Calculate $\sum x$, $\sum y$, $\sum xy$, and $\sum x^2$.
- 3 Calculate the slope:

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

- 4 Calculate the intercept:

$$a = \frac{\sum y - b \sum x}{n}$$

Example:

Input:

$n = 4$

$x: 1 \ 2 \ 3 \ 4$

$y: 2 \ 4 \ 5 \ 8$

Prediction $x = 5$

Expected Output:

Slope = 1.9

Intercept = 0.5

Predicted $y = 10.0$

Question 26: Algorithm and Example



Algorithm

5 Read the value of x to be predicted.

6 Calculate:

$$y = a + bx$$

7 Display a , b , and predicted y .

Task

Implement the complete algorithm in C using user-defined functions.
Do not use any library function for regression.

Thank You!

Questions?